

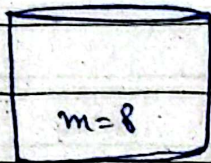
0/1 Knapsack Problem

Item	1	2	3	4
Profit P	1	2	5	6
Weight W	2	3	4	5

$$m = 8$$

$$n = 4$$

$x_i = 0/1$ can be 0 or 1.



Aim ~ Maximize Profits (Optimization Problem)

total possibilities =

0	0	0	0
1	1	1	1
0	0	0	1
0	0	1	1
⋮			

$$2^4 = 16$$

total possibilities	2^n
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$$m = 8$$

$n = 4 \rightarrow$ no. of items/objects

$V(i, w)$			0	1	2	3	4	5	6	7	8
Profit wt (i)			0	0	0	0	0	0	0	0	0
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	(2)	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5	6	6	7	(8)

Formula:-

$$V[i, w] = \max \{ V[i-1, w], P[i] + V[i-1, w - w_i] \}$$

Working for 4th row:-

$$V[4, 1] = \max (V(3, 1), 6 + V(3, w-5))$$

$$\downarrow \quad \quad \quad 6 + V(3, 1-5) = V(3, -4) \text{ // invalid}$$

$$V(3, 1) = (0)$$

$$V[4, 2] = V(3, 2), 6 + V(3, 2-5)$$

$$\downarrow \quad \quad \quad \text{invalid}$$

$$V[4, 3] = V(3, 3), 6 + V(3, 3-5)$$

$$\downarrow \quad \quad \quad \text{invalid}$$

$$V[4, 4] = V(3, 4), 6 + V(3, 4-5)$$

$$\downarrow \quad \quad \quad \text{invalid}$$

$$V[4, 5] = \max (V(3, 5), 6 + V(3, 0))$$

$$= \max (5, 6) = (6)$$

$$V[4, 6] = \max (V(3, 6), 6 + V(3, 1))$$

$$= \max (6, 6) = (6)$$

$$V[4, 7] = \max (V(3, 7), 6 + V(3, 2))$$

$$= \max (7, 7) = (7)$$

$$V[4, 8] = \max (V(3, 8), 6 + V(3, 3))$$

$$V[4, 8] = \max (7, 8) = (8)$$

Maximum Profit = 8

Profit is 8 when 4th object

$$V(4, 8) = 8$$

6i

Pi

$$8 - 6 = 2$$

↓

Go to 2nd row → 2

$$2 - 2 = 0$$

$x_1 \quad x_2 \quad x_3 \quad x_4$

$$\{ 0 \quad 1 \quad 0 \quad 1 \}$$

Include x_2 and x_4

Matrix Chain Multiplication

$$A_1 \times A_2 \times A_3 \times A_4 \times A_5$$

$$\begin{matrix} 4 \times 10 & 10 \times 3 & 3 \times 12 & 12 \times 20 & 20 \times 7 \\ p_1 p_2 & p_2 p_3 & p_3 p_4 & p_4 p_5 & p_5 p_6 \end{matrix}$$

$$4 \times 10 \times 3 + 4 \times 3 \times 12 + 4 \times 12 \times 20 + 4 \times 20 \times 7 = 1284 \text{ Multiplication Operations}$$

VERY LARGE COMPUTATIONAL TIMES !!!

① Check if problem has optimal substructure

$$\begin{matrix} (A_1 & A_2 & A_3) & (A_4 & A_5) \\ i & & k & k+1 & j \end{matrix}$$

$$M[i, j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{ M[i, k] + M[k+1, j] + p_i - p_k p_j \} \end{cases}$$

i \ j	1	2	3	4	5
1	0	120	264	1080	1374
2	X	0	360	1320	1350
3	X	X	0	720	1140
4	X	X	X	0	1680
5	X	X	X	X	0

To compute $M[1,2]$

$$M[1,2] = \min_{1 \leq k < 2} \{M[1,1] + M[1+1,2] + p_1 p_1 p_2\}$$

$$M[1,2] = \min_{1 \leq k < 2} \{0 + 0 + 4 \times 10 \times 3\}$$

$$M[1,2] = 120$$

$$M[2,3] = \min_{2 \leq k < 3} \{M[2,2] + M[2+1,3] + p_1 p_2 p_3\}$$

$$M[2,3] = \min_{2 \leq k < 3} \{0 + 0 + 3 \times 12 \times 20\}$$

$$M[2,3] = 360$$

$$M[3,4] = \min_{3 \leq k < 4} \{M[3,3] + M[3+1,4] + p_2 p_3 p_4\}$$

$$= 3 \times 12 \times 20$$

$$= 720$$

$$M[4,5] = \min_{4 \leq k < 5} \{M[4,4] + M[4+1,5] + p_3 p_4 p_5\}$$

$$= 12 \times 20 \times 7$$

$$= 1680$$

$$M[1,3] = \min_{1 \leq k < 3} \{M[1,1] + M[2,3] + p_0 p_1 p_3\},$$

where k can be 1, 2

$$M[1,2] + M[3,3] + p_0 p_2 p_3\}$$

$$= \min \{0 + 360 + (4 \times 10 \times 12), 120 + 0 + (4 \times 3 \times 12)\}$$

$$= \min \{840, 264\}$$

$$= 264$$

$$M[2,4] = \min_{2 \leq k < 4} \{ M[2,2] + M[3,4] + p_2 p_3 p_4, \\ \downarrow$$

here k can be 2,3

$$M[2,3] + M[4,4] + p_2 p_3 p_4 \}$$

$$= \min \{ 0 + 720 + (10 \times 3 \times 20), 360 + 0 + (10 \times 12 \times 20) \}$$

$$= \min \{ 1320, 2760 \}$$

$$= 1320$$

$$M[3,5] = \min_{3 \leq k < 5} \{ M[3,3] + M[4,5] + p_2 p_3 p_5, \\ \text{here k can be 3,4}$$

$$M[3,4] + M[5,5] + p_2 p_4 p_5 \}$$

$$= \min \{ 0 + 1680 + (3 \times 12 \times 7), 720 + 0 + (3 \times 20 \times 7) \}$$

$$= \min \{ 1932, 1140 \}$$

$$= 1140$$

$$M[1,4] \rightarrow \begin{matrix} 1 \leq k < 4 \\ 1 \leq k < 4 \end{matrix}$$

here k can be 1,2,3

$$= \min \{ M[1,1] + M[2,4] + p_0 p_1 p_4, \\$$

$$M[1,2] + M[3,4] + p_0 p_2 p_4, \\$$

$$M[1,3] + M[4,4] + p_0 p_3 p_4 \}$$

$$= \min \{ 0 + 1320 + (4 \times 10 \times 20), 120 + 720 + (4 \times 3 \times 20), \\ 264 + 0 + (4 \times 12 \times 20) \}$$

$$= \min \{ 2120, 1080, 1224 \}$$

$$= 1080$$

$$M[i, j] = \begin{cases} 0, & \text{if } i=j \\ \min_{i \leq k < j} \{ M[i, k] + M[k+1, j] + p_{i-1} p_k p_j \} \end{cases}$$

Date: _____

$M[1, 5]$

where k can be 1, 2, 3, 4

$$= \min_{1 \leq k < 5} \{ M[1, 1] + M[2, 5] + p_0 p_1 p_5, \\ M[1, 2] + M[3, 5] + p_0 p_2 p_5, \\ M[1, 3] + M[4, 5] + p_0 p_3 p_5, \\ M[1, 4] + M[5, 5] + p_0 p_4 p_5 \}$$

$$= \min \{ 0 + M[2, 5] + (4 \times 10 \times 7), \\ 120 + 1140 + (4 \times 3 \times 7), \\ 264 + 1680 + (4 \times 3 \times 7), \\ 1080 + 0 + (4 \times 20 \times 7) \}$$

$$= \min (M[2, 5] + 280, 1344, 2028, 1640) \\ = \min (1350 + 280, 1344, 2028, 1640) = \min (1630, 1344, 2028, 1640) \\ = 1344$$

$M[2, 5]$

where k can be 2, 3, 4

$$= \min_{2 \leq k < 5} \{ M[2, 2] + M[3, 5] + p_1 p_2 p_5, \\ M[2, 3] + M[4, 5] + p_1 p_3 p_5, \\ M[2, 4] + M[5, 5] + p_1 p_4 p_5 \}$$

$$= \min (0 + 1140 + (10 \times 3 \times 7), 360 + 1680 + (10 \times 12 \times 7), \\ 1320 + 0 + (10 \times 20 \times 7))$$

$$= \min (1350, 2880, 2720)$$

$$= 1350$$

0/1 KNAPSACK PRACTICE PROBLEM

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Benefit	15	35	10	9	9
Weight	4	12	4	4	5

maximum, total weight in the knapsack is $W=9$.
allowable

$$W=9$$

B	w	w →										
		0	0	1	2	3	4	5	6	7	8	9
15	4	0	0	0	0	0	0	0	0	0	0	0
35	12	2	0	0	0	0	15	15	15	15	15	15
10	4	3	0	0	0	0	15	15	15	15	25	25
9	4	4	0	0	0	0	15	15	15	15	25	25
9	5	5	0	0	0	0	15	15	15	15	25	25

Formula:-

$$V[i, w] = \max \{ V[i-1, w], B[i] + V[i-1, w-w_i] \}$$

$$V[1, w] = \max \{ V[0, w], 15 + V[0, w-4] \}$$

$$\begin{aligned} & \downarrow V[0, -4], V[0, -3], V[0, -2], V[0, -1] \\ & \text{invalid} \\ & \begin{array}{l} V[0, 0] \\ V[0, 1] \\ V[0, 2] \\ \vdots \\ V[0, 4] \end{array} = 0 \quad \begin{array}{l} 15 + V[0, 4-4] \\ = \max(0, 15+0) \end{array} \end{aligned}$$

$$\text{for } V[1, 5] = \max \{ V[0, 5], 15 + V[0, 1] \} = \max(0, 15+0) = 15$$

$$\text{for } V[1, 6] = \max \{ V[0, 6], 15 + V[0, 2] \} = \max(0, 15+0) = 15$$

$$\text{for } V[1, 7] = \max \{ V[0, 7], 15 + V[0, 3] \} = \max(0, 15+0) = 15$$

$$\text{for } V[1, 8] = \max \{ V[0, 8], 15 + V[0, 4] \} = \max(0, 15+0) = 15$$

$$\text{for } V[1, 9] = \max \{ V[0, 9], 15 + V[0, 5] \} = \max(0, 15+0) = 15$$

$$V[2, w] = \max \{ V[1, w], 35 + V[1, w-12] \}$$

for w ranging 0 → 9

$V[1, 0-12]$
 $V[1, 9-12]$ invalid so

$$V[2, 0] = V[1, 0] = 0$$

$35 + V[1, w-12]$ leave this

$$V[2,1] = V[1,1] = 0$$

$$V[2,2] = V[1,2] = 0$$

$$V[2,3] = V[1,3] = 0$$

$$V[2,4] = V[1,4] = 15$$

$$V[2,5] = V[1,5] = 15$$

$$V[2,6] = V[1,6] = 15$$

$$V[2,7] = V[1,7] = 15$$

$$V[2,8] = V[1,8] = 15$$

$$V[2,9] = V[1,9] = 15$$

$$V[3,w] = \max\{V[2,w], 10 + V[2,w-4]\}$$

$$V[3,0] = V[2,0]$$

$\downarrow 0-3 \rightarrow \text{invalid}$

$$V[3,1] = V[2,1]$$

$$10 + V[2,4-4]$$

$$V[3,2] = V[2,2]$$

$$10 + V[2,0]$$

$$V[3,3] = V[2,3]$$

$$V[3,4] = \max\{V[2,4], 10 + V[2,0]\}$$

$$= \max\{15, 10 + 0\} = 15$$

$$V[3,5] = \max\{V[2,5], 10 + V[2,1]\} = 15$$

$$V[3,6] = \max\{V[2,6], 10 + V[2,2]\} = 15$$

$$V[3,7] = \max\{V[2,7], 10 + V[2,3]\} = 15$$

$$V[3,8] = \max\{V[2,8], 10 + V[2,4]\} = \max(15, 15+10) = 25$$

$$V[3,9] = \max\{V[2,9], 10 + V[2,5]\} = \max(15, 15+10) = 25$$

$$V[4,w] = \max\{V[3,w], 9 + V[3,w-4]\}$$

$$V[4,0] = \max\{V[3,0], 9 + V[3, \downarrow 0-3 \rightarrow \text{invalid}]\}$$

$$V[4,1] = V[3,1]$$

$$V[4,2] = V[3,2]$$

$$V[4,3] = V[3,3]$$

$$V[4,4] = \max(15, 9+0) = 15$$

$$V[4,5] = \max(15, 9+0) = 15$$

$$V[4,6] = \max(15, 9+0) = 15$$

$$V[4,7] = \max(15, 9+0) = 15$$

$$V[4,8] = \max(25, 9+15) = 25$$

$$V[4,9] = \max(25, 9+15) = 25$$

$$V[S,w] = \max\{V[4,w], 9+V[4,w-5]\}$$

↳ $w \rightarrow 0-4$ invalid

$$V[S,0] = V[4,0] = 0$$

$$V[S,1] = V[4,1] = 0$$

$$V[S,2] = V[4,2] = 0$$

$$V[S,3] = V[4,3] = 0$$

$$V[S,4] = V[4,4] = 15$$

$$V[S,5] = \max\{V[4,5], 9+V[4,0]\}$$

$$= \max(15, 9+0) = 15$$

$$V[S,6] = \max\{V[4,6], 9+V[4,1]\}$$

$$= \max(15, 9+0) = 15$$

$$V[S,7] = \max\{V[4,7], 9+V[4,2]\}$$

$$= \max(15, 9+0) = 15$$

$$V[S,8] = \max\{V[4,8], 9+V[4,3]\}$$

$$= \max(25, 9+0) = 25$$

$$V[S,9] = \max\{V[4,9], 9+V[4,4]\}$$

$$= \max(25, 9+15) = 25$$